Concordant or not? Performance and concordance rate analysis of ten prediction algorithms on clinically relevant variants from the BRCA1 and BRCA2 genes

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SCAN ME

INTRODUCTION

The rapid integration of next-generation sequencing into clinical genetics, particularly in the screening of the BRCA1 and BRCA2 genes, has led to the detection of thousands of missense variants with unknown clinical significance. Due to their uncertain impact on human health, these variants are commonly omitted from medical records. To address this issue and to characterize the function of these variants, a multitude of machine learning-based algorithms have been developed over the past decade. Even so, current guidelines encourage the use of multiple algorithms as opposed to just one, advising that the computational analysis should be considered as supporting evidence only if the used algorithms are in concordance with one another[1].

AIMS

To evaluate the performance and analyze the concordance rates of ten widely used in-silico tools (PMut, Panther-PSEP, PolyPhen-2-HumDiv, PolyPhen-2-HumVar, SNPs&GO, PROVEAN, PhD-SNP, SIFT, and META-SNP)* on missense variants from BRCA1 and BRCA2 genes.

METHODS

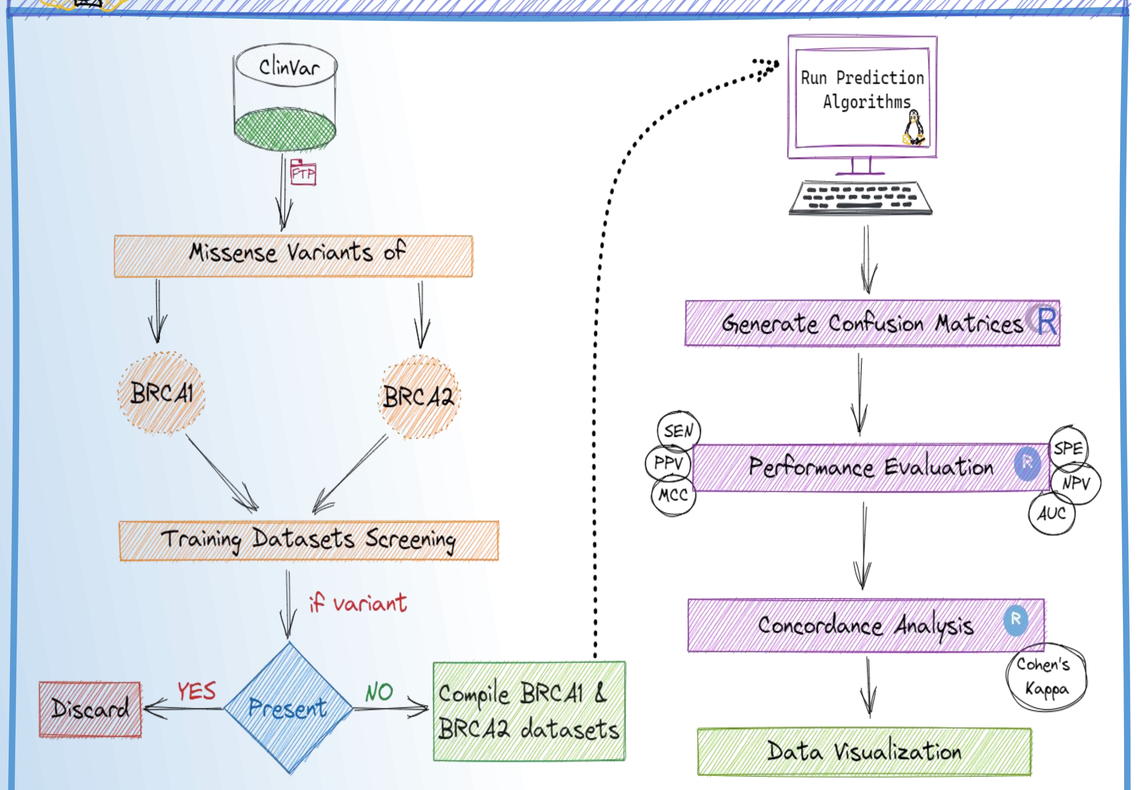

Figure 1: *Schematic representation of the workflow, from dataset generation to data visualization. SEN: Sensitivity; SPE: Specificity; PPV: Positive Predictive Value; NPV: Negative Predictive Value; MCC: Matthews Correlation Coefficient; AUC: Area Under the Curve. The flowchart was drawn using Excalidraw.*

CONCLUSIONS

- The performance of the prediction algorithms across all the evaluated statistical metrics differed substantially between the two genes.
- The use of PROVEAN to prioritize BRCA1 and BRCA2 variants may result in the erroneous classification of pathogenic variants, ultimately hindering functional studies.
- Overall, pairings composed of best-performing prediction algorithms led to an increase in true concordance instances between the algorithms.
- Utilizing multiple prediction algorithms, as advised by current guidelines, increases the likelihood of discordance between them, rendering the computational analysis ineffective.

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RESULTS

I. According to Matthews Correlation Coefficient (MCC) values, SNPs&GO and PMut were the most efficient algorithms for BRCA1 and BRCA2, respectively (Figure 2A).

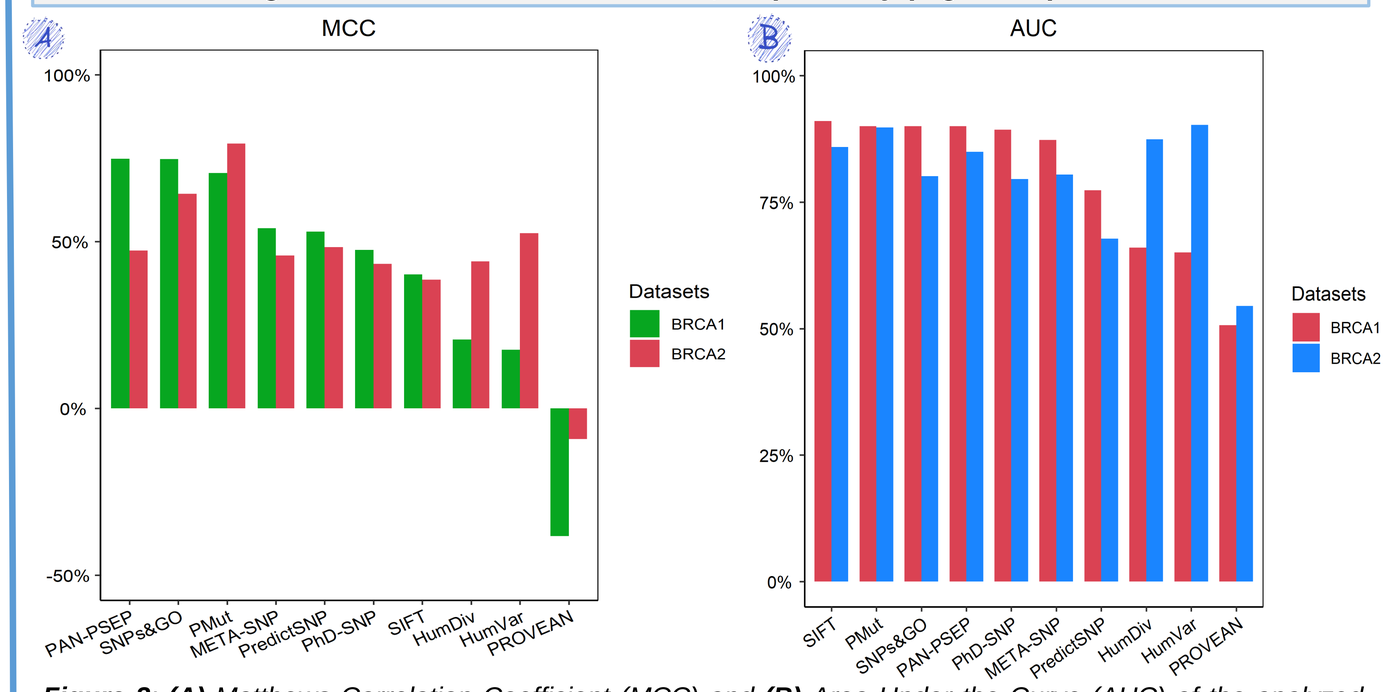

Figure 2*: **(A)** Matthews Correlation Coefficient (MCC) and **(B)** Area Under the Curve (AUC) of the analyzed prediction algorithms on the BRCA1 and BRCA2 datasets[2].*

II. The PROVEAN algorithm was unable to correctly classify the majority of pathogenic missense variants in both genes, displaying the poorest performance among the evaluated algorithms (Figure 2B, 2C, 2D).

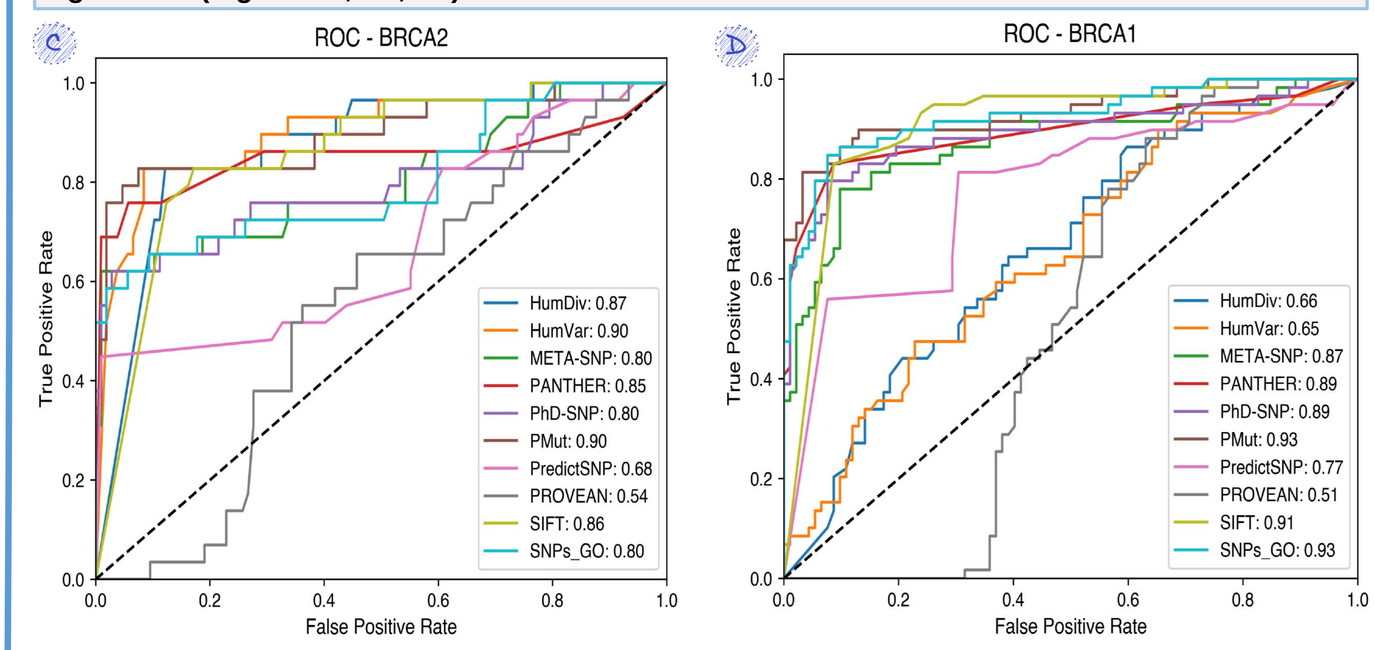

Figure 2*: Area Under the Curve (AUC) and the respective values of the prediction algorithms in the BRCA1 **(C)** and BRCA2 **(D)** datasets[2].*

III. The best-performing prediction algorithms showed higher concordance rates between each other as opposed to other algorithm pairs. PROVEAN displayed remarkably low concordance rates across all the calculated algorithm pairs (Figure 2F, 2E).

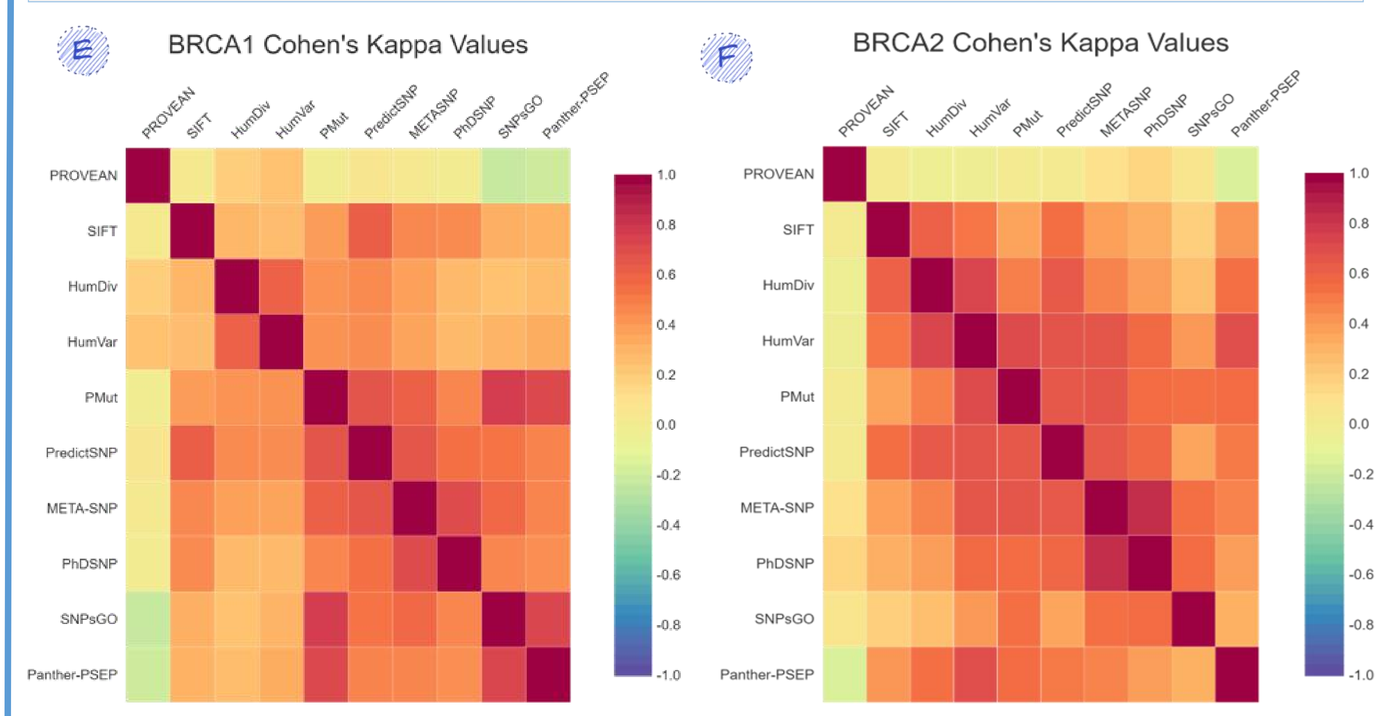

Figure 2*: Pairwise heatmaps of the Cohen's Kappa values for pair of algorithms in the BRCA1 dataset **(E)** and BRCA2 dataset **(F)**. Cohen's Kappa was utilized to determine the concordance rate between algorithms*

REFERENCES

1. Richards, Sue, *et al*. "Standards and Guidelines for the Interpretation of Sequence Variants: A Joint Consensus Recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology." ***Genetics in Medicine***.
2. Qorri E, *et al*. "A Comprehensive Evaluation of the Performance of Prediction Algorithms on Clinically Relevant Missense Variants". ***International Journal of Molecular Science***.

Reference list of the analyzed prediction algorithms is accessible via the adjacent QR code.